UAPOML Week 1

Pallav Pratibh

Q1. Returns from prices.
Prices: $100, 108, 103, 115, 110$.

(a) Simple returns $R_t = (P_t - P_{t-1})/P_{t-1}$:

$$R_1 = 8.00\%, \quad R_2 = -4.63\%, \quad R_3 = 11.65\%, \quad R_4 = -4.35\%.$$

(b) Log returns $r_t = \ln(P_t/P_{t-1})$:

$$r_1 = 7.70\%, \quad r_2 = -4.74\%, \quad r_3 = 11.02\%, \quad r_4 = -4.45\%.$$

(c) The two agree closely. Looking at the gaps $|R_t - r_t|$, the approximation is **least accurate at Month 3** ($\approx$ 0.63 percentage points). This follows from the expansion $\ln(1+R) = R - \frac{R^2}{2} + \cdots$; the leading error term is $R^2/2$, which grows with $|R|$. Month 3 has the largest move (+11.65%), so it has the largest error.

Q2. Expected portfolio return.

(a) Linearity of expectation states $E[aX + bY] = aE[X] + bE[Y]$ for constants a, b. Applying it to $R_p = w_1R_1 + w_2R_2$:

$$E[R_p] = w_1E[R_1] + w_2E[R_2] = 0.4(0.15) + 0.6(0.07) = \mathbf{10.2}\%.$$

(b) **No.** Linearity of expectation holds for *any* random variables, correlated or not. Correlation only affects the *variance* of the portfolio, never its mean.

Q3. Two-asset variance, derived.
Let $R_p = w_1R_1 + w_2R_2$. Using $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$ with $X = w_1R_1$, $Y = w_2R_2$:

$$\begin{aligned}\sigma_p^2 &= \mathrm{Var}(w_1R_1) + \mathrm{Var}(w_2R_2) + 2\,\mathrm{Cov}(w_1R_1,\, w_2R_2)\\ &= w_1^2\,\mathrm{Var}(R_1) + w_2^2\,\mathrm{Var}(R_2) + 2w_1w_2\,\mathrm{Cov}(R_1, R_2)\\ &= w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\,\mathrm{Cov}(R_1, R_2).\end{aligned}$$

Constants pull out of variance squared and out of covariance linearly. Finally substitute $\mathrm{Cov}(R_1, R_2) = \rho_{12}\sigma_1\sigma_2$:

$$\boxed{\sigma_p^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{12}\sigma_1\sigma_2}$$

Q4. Risk vs. correlation.
$\sigma_A = 25\%$, $\sigma_B = 12\%$, $w_A = w_B = 0.5$. Here $\sigma_p^2 = 0.019225 + 0.015\,\rho$.

(a) $\rho = +1: \ \sigma_p = \mathbf{18.50}\%, \quad \rho = 0: \ \sigma_p = \mathbf{13.87}\%, \quad \rho = -1: \ \sigma_p = \mathbf{6.50}\%.$

(b) Only at $\rho = +1$. There $\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_Aw_B\sigma_A\sigma_B = (w_A\sigma_A + w_B\sigma_B)^2$, a perfect square, so $\sigma_p = w_A\sigma_A + w_B\sigma_B$ — exactly the weighted average.

(c) At $\rho = -1$, $\sigma_p = |w_A\sigma_A - w_B\sigma_B|$. Setting this to zero gives $w_A\sigma_A = (1 - w_A)\sigma_B$, so

$$w_A = \frac{\sigma_B}{\sigma_A + \sigma_B} = \frac{0.12}{0.37} = \mathbf{0.324}.$$

Q5. Why $\rho < 1$ helps (<150 words).

The whole effect lives in the cross term $2w_1w_2\rho\sigma_1\sigma_2$ of the variance formula. At $\rho = 1$ this term is as large as possible, the variance becomes a perfect square, and σ_p equals the weighted average $w_1\sigma_1 + w_2\sigma_2$. As ρ drops below 1, the cross term shrinks, the total variance falls, and σ_p slips *below* that weighted average. Intuitively, when two assets aren't perfectly in sync, a dip in one is partly offset by the other, so part of the swings cancel and the combined return is smoother. The lower the correlation, the more cancellation, and the bigger the diversification benefit — which is why $\rho = -1$ can even drive risk to zero.

Q6. Covariance matrix.

Using $\Sigma_{ij} = \rho_{ij}\sigma_i\sigma_j$ with $\sigma = (0.25, 0.12, 0.04)$:

(a)

$$\Sigma = \begin{pmatrix} 0.0625 & 0.0120 & 0.00100 \\ 0.0120 & 0.0144 & 0.00096 \\ 0.00100 & 0.00096 & 0.0016 \end{pmatrix}.$$

(b) Symmetric because $\Sigma_{ij} = \rho_{ij}\sigma_i\sigma_j = \rho_{ji}\sigma_j\sigma_i = \Sigma_{ji}$ (correlation is symmetric, $\rho_{ij} = \rho_{ji}$). The off-diagonals mirror across the diagonal.

Q7. Sharpe ratios. $R_f = 4\%, \ S = (E[R_p] - R_f)/\sigma_p.$

(a) $S_X = \dfrac{10}{22} = \mathbf{0.455}, \quad S_Y = \dfrac{5}{10} = \mathbf{0.500}, \quad S_Z = \dfrac{16}{35} = \mathbf{0.457}.$

(b) Ranking: **Y** > **Z** > **X**. A mean–variance investor prefers **Y**: it delivers the most excess return per unit of risk.

(c) Z wins on raw return but is the least efficient. An investor who wants Z's high return can instead *lever* the more efficient portfolio Y (borrow at R_f and scale up). Moving along Y's capital allocation line gives a higher return than Z at the same risk, so a rational investor still prefers Y.

Q8. GMV, the frontier, and the CML.

(a) The **Global Minimum Variance (GMV)** portfolio is the feasible portfolio with the smallest possible variance. Since the horizontal axis of the frontier is risk σ_p, "smallest variance" means "smallest σ_p," which is the **leftmost point**.

(b) **Not correct.** The lower half of the minimum-variance frontier is *inefficient*: for every point there, a point directly above on the upper half has the *same risk* but a *higher return*. A low-risk investor should sit at the GMV point or move up the efficient (upper) frontier — never on the lower half.

(c) The **Capital Market Line** is drawn from the risk-free rate R_f on the return axis, straight up to where it just touches (is tangent to) the efficient frontier — the *tangency / market portfolio*. Points on it mix the risk-free asset with that market portfolio. Its **slope** is $\left(E[R_m] - R_f\right)/\sigma_m$, the market Sharpe ratio — the extra return the market pays per unit of risk (the "price of risk").

Q9. Why MVO inputs are hard.

(a) Standard error of the sample mean: $\text{SE}(\hat{\mu}_i) = \sigma_i/\sqrt{n} = 0.30/\sqrt{36} = \mathbf{5}\%$. That 5% error band is enormous next to the return we're trying to estimate, so a mean built from history alone is extremely noisy and unreliable.

(b) **Error maximisation:** MVO treats $\hat{\mu}$ as if it were exact, so it loads heavily onto assets whose returns were *over*estimated and shorts those *under*estimated — it actively amplifies estimation error into extreme, unstable weights. Replacing $\hat{\mu}$ with **ML-predicted returns** (which use more information and are regularised) gives steadier, more accurate inputs, so there is less error for MVO to blow up.

(c) For $N = 50$, Σ has $\dfrac{N(N+1)}{2} = \dfrac{50 \cdot 51}{2} = \mathbf{1275}$ unique entries. Estimating that many parameters from limited data gives noisy, possibly ill-conditioned estimates — the matrix can be unstable or nearly non-invertible, which wrecks the optimisation.